

Chapter 6 — Magnetic Resonance Imaging

INTRODUCTION

Position relative to computed tomography

- **Computed tomography (CT) remains the mainstay** of urological cross-sectional imaging, but the resolution gap between CT and magnetic resonance imaging (MRI) is narrowing.
- The central advantage of MRI is **excellent soft-tissue contrast resolution**, in many situations without intravenous contrast.

How the image is generated

- **Sequence:** patient on a gantry through the magnet bore → free water protons align along the **z-axis** (head-to-toe, through the bore) → a radiofrequency (RF) coil over the region transmits RF pulses → when the pulse stops, protons **release energy**, which is detected and processed into the image.
- A pulse sequence exploits the different ways tissues absorb and release this energy. Most coils now transmit and receive (**dual-channel RF**).

Weighting

- **T1 weighting** reflects the time to return to equilibrium along the z-axis (longitudinal relaxation); **T2 weighting** reflects the time to return to equilibrium in the x-y plane (transverse relaxation).
- **Fluid is dark on T1 and bright on T2.**
- In the kidney, **the cortex is brighter than the medulla on T1**; on T2 the medulla is relatively brighter.

Advantages of MRI

- **No ionising radiation**, and therefore no risk of radiation-induced second malignancy.
- **Modality of choice in pregnancy, renal insufficiency and iodinated contrast allergy.**

Gadolinium-based contrast agents and nephrogenic systemic fibrosis

- Gadolinium-based contrast agents (GBCAs) are **non-iodinated**. Iodinated CT contrast absorbs X-rays; **GBCAs shorten the relaxation times of water**, raising signal (enhancement), usually assessed on T1.
- Gadolinium is a toxic heavy metal; it is **chelated** to prevent cellular uptake, making it non-toxic in almost all patients with normal renal function.
- **Nephrogenic systemic fibrosis (NSF):** the historical concern with glomerular filtration rate (GFR) **below 30 mL/min/1.73 m²** — gadolinium deposits in skin and muscle as an insoluble precipitate and causes systemic fibrosis. This led to gadolinium being avoided in severe renal dysfunction.
- GBCAs are not all alike. The American College of Radiology (ACR) states that **group II agents carry little to no NSF risk, even in end-stage renal disease:** gadobenate dimeglumine, gadoteridol, gadoterate meglumine and gadobutrol.

Concept — MRI trades CT's speed and spatial resolution for tissue contrast without radiation, and the historical contrast barrier in renal failure has largely fallen with group II agents.

ADRENAL MAGNETIC RESONANCE IMAGING

- MRI characterises tissue without contrast; **minute quantities of lipid separate benign adenoma from malignancy.** Most adrenal masses are incidental and non-functioning.
- Adenomas are usually **under 3 cm and non-functional**, with a **high lipid content** that makes them readily distinguishable from malignant lesions.

- Three lipid techniques exploit the different behaviour of fat and water protons: inversion recovery, **chemical shift imaging (CSI)** and fat saturation. **CSI is the commonest in urology.** Spectral presaturation with inversion recovery (SPIR) and spectral presaturation attenuated inversion recovery (SPAIR) detect fat by selective excitation of fat protons and T1 relaxation.

Adrenal Adenoma

- **Physics of CSI:** in water, oxygen pulls the electron cloud away from hydrogen; in fat ($-\text{CH}_2-$) the less electronegative carbon has a smaller effect. The resulting difference in shielding gives **different precession frequencies — the chemical shift.**
- **In-phase (IP):** echo time (TE) set so fat and water signals **add** within a voxel. **Out-of-phase (OP):** TE set so signals **cancel**, so a voxel containing both fat and water loses signal.
- **Signal loss on OP relative to IP = intracytoplasmic (microscopic) fat = adenoma.** No change makes fat less likely.
- **Qualitative (visual) comparison is sensitive but poorly specific;** the quantitative signal intensity (SI) index raises specificity to very high levels:
 - **SI index (%) = (SI in-phase – SI out-of-phase) / SI in-phase × 100**
 - **No agreed threshold;** reported cut-offs range from **1.7% to 20%**, with 5% at 3 tesla performing best in one series.
- **Lipid-poor adenomas (10–30%)** give indeterminate CSI. Adrenocortical carcinoma washes out slowly, so **contrast-enhanced CT with washout may better separate lipid-poor adenoma** from other masses.
- In unilateral functioning adenoma, CT and MRI are limited for lateralisation; **adrenal vein sampling** is a useful adjunct.
- **Figure 6.1:** IP T1 shows a left adrenal mass isointense to muscle; OP T1 shows **signal drop-out**; single-shot T2 shows a low-signal nodule.

Adrenal Cortical Carcinoma

- Diagnosis combines clinical and imaging features. **About 62% are hormonally active.**
- **Risk rises with size:** lesions ≤ 4 cm account for 2% of adrenocortical carcinomas (ACCs); 4–6 cm, 6%; over 6 cm, 25%.
- **T2 heterogeneous and high; T1 with gadolinium heterogeneous enhancement; CSI low signal** (no lipid drop-out).
- **Local venous thrombosis** is associated with ACC and is detectable on MRI.
- ACC is metabolically active: **fluorodeoxyglucose-positron emission tomography (FDG-PET)** separates ACC from adenoma with high sensitivity and specificity.
- **Figure 6.2:** hyperintense on moderately T2-weighted short tau inversion recovery (STIR) but isointense on heavily T2-weighted single-shot fast spin echo — **apparent T2 brightness depends on the degree of T2 weighting.**

Myelolipoma

- Benign mass of **mature fat and bone marrow elements**; about **6.5% of incidentally detected adrenal masses.** Lesions over 4 cm can mimic malignancy.
- **High T1 signal; signal suppression on frequency-selective fat suppression** (macroscopic fat); **India ink artefact** — a dark line around the lesion or at organ interfaces on OP images, from voxels containing both fat and water.

- **Figure 6.3:** signal isointense to abdominal fat on T2 and IP T1; **no drop on OP relative to IP** (gross fat, not intracytoplasmic lipid mixed with water); loss of signal on fat suppression.

Metastasis

- Suspect in the setting of a **known primary**. Large, irregular, heterogeneous, occasionally necrotic.
- **High T2 signal** from higher fluid content than adenoma; heterogeneous gadolinium enhancement with **delayed peak enhancement** relative to adenoma; a time-to-peak cut-off of about **53 seconds or more** characterises metastasis well.
- **Trap — lipid-containing primaries:** a metastasis carries the histology of its primary, so it can be **CSI-positive and mimic adenoma**. Reported with **clear cell renal cell carcinoma, liposarcoma and hepatocellular carcinoma**.
- **Figure 6.4:** metachronous right adrenal metastasis after nephrectomy for clear cell carcinoma — **drop-out on OP** (microscopic fat) and T2 hyperintensity.

Pheochromocytoma

- High T2 signal (the "light bulb" sign) was traditionally considered diagnostic, but **about a third of pheochromocytomas are T2-dark**.
- **Pheochromocytoma, ACC and metastasis can all be T2-bright**, and brightness varies with T2 weighting — the sign is neither sensitive nor specific.
- **Figure 6.5:** isointense on heavily weighted T2; bright on moderately weighted fat-suppressed T2; **marked early enhancement** on post-contrast T1.

Haemorrhage and Other Pathologies

- **Adrenal haematoma** changes with age as red cells break down and haemosiderin is deposited: early **isointense to hypointense on T1 and low on T2** → at **1–7 weeks hyperintense on T1 fat-suppressed and T2**; a **low-signal haemosiderin rim on both T1 and T2**.
- **Teratoma:** mixed tissues (bone, teeth, hair, muscle) give **mixed T1 and T2 signal**; enhancement of the peripheral rim, solid components and internal septations.
- **Intermediate SI index on CSI:** lymphoma, neuroblastoma, ganglioneuroma, haemangioma and granulomatous disease.

TABLE 6.1 MORPHOLOGIC AND IMAGING CHARACTERISTICS OF INCIDENTAL ADRENAL LESIONS (REBUILT)

Lesion	Size	Shape / texture	Unenhanced CT (HU)	15-min CT washout	MRI
Adenoma	1–4 cm	Smooth, round; homogeneous	<10 in 70%	RPW >40%; APW >60%	SI drop on OP
Myelolipoma	1–5 cm	Smooth, round; macroscopic fat	<0, often < -50	No data	High T1; India ink; variable OP drop
Cyst	Variable	Smooth, round	<10	Does not enhance	High T2
Metastasis	Variable	Heterogeneous when large	>10	RPW <40%	High T2
ACC	>4 cm	Variable	>10	RPW <40%	Intermediate to high T2
Pheochromocytoma	Variable	Variable	>10, rarely <10	RPW <40%	High T2
Lymphoma	Variable	Variable	>10	RPW <40%	Intermediate SI

Lesion	Size	Shape / texture	Unenhanced CT (HU)	15-min CT washout	MRI
Neuroblastoma	Variable	Smooth, round	>10	RPW <40%	Variable if necrotic
Haematoma	Variable	Smooth; variable	>10, sometimes >50	No data	Variable
Ganglioneuroma, haemangioma	Variable	Variable	>10	No data	Usually intermediate SI
Granulomatous	1–5 cm	Smooth; usually homogeneous	>10	No data	Usually intermediate SI

HU, Hounsfield units; RPW, relative percentage washout; APW, absolute percentage washout. Data from Boland GW, Blake MA: Incidental adrenal lesions, Radiology 2008.

- **What the table is for:** it lines up the two tests that clear an incidentaloma — **low attenuation or high washout on CT, and OP drop-out on MRI**. Only adenoma passes both.
- **Pattern:** everything that is not adenoma or myelolipoma is **>10 HU with RPW <40%**; the MRI column then separates by T2 signal, which overlaps.
- **Trap:** a high T2 signal does not identify pheochromocytoma — metastasis, ACC and cyst share it.

Concept — Adrenal MRI hinges on fat: microscopic fat gives OP drop-out and means adenoma; gross fat gives fat-suppression loss and India ink, without OP drop, and means myelolipoma. T2 brightness is non-specific; lipid-poor adenomas go to CT washout, and lipid-rich metastases are the trap.

RENAL MAGNETIC RESONANCE IMAGING

Cysts

- **Simple cysts** look alike on ultrasound, CT and MRI; **MRI characterises complex cysts better**.
- **Haemorrhagic cyst contents are T1-bright** (paramagnetic blood-breakdown products — methaemoglobin); **proteinaceous contents are also T1-bright**; **chronic haemorrhage gives a black ring on T2** along the wall.
- **Benign cysts, including benign complex cysts, do not enhance.**
- As an independent risk factor for renal cell carcinoma (RCC), **wall enhancement is more sensitive and specific than wall calcification**. On CT, calcification can cause artefact that hides enhancing mural nodules; **on MRI calcium is a signal void and does not interfere**, so MRI detects enhancement in a calcified complex cyst better than CT.
- **Pseudocapsule:** a low-signal rim on T1 and T2, assessed better on MRI than CT. **Absence of the pseudocapsule predicts pT3a disease.**

Subtyping renal cell carcinoma

- **Multiparametric approach:** T1; multiplanar T2 with and without fat suppression; dynamic contrast-enhanced (DCE) phases (arterial, corticomedullary, nephrographic, excretory); diffusion-weighted imaging (DWI) with apparent diffusion coefficient (ADC) maps; and CSI. This subtypes renal masses **better than CT**.
- **Clear cell RCC (ccRCC)**, the commonest subtype: **heterogeneous high T2 signal** from haemorrhage, necrosis and cysts (Fig. 6.6).
- **Papillary RCC (pRCC): homogeneous low T2 signal** from haemosiderin in histiocytes (Fig. 6.7). A haemorrhagic cyst with an enhancing peripheral wall, or a solid hypo-enhancing T2-low mass, separates pRCC from other subtypes with high specificity.

- **DWI** detects restricted water movement in intracellular and extracellular spaces; **ADC $\geq 1281 \times 10^{-6}$ mm²/s favours clear cell over non-clear cell RCC.**
- **Oncocytoma's central scar is T2-bright**, but present in only **54–80%**, and it is also seen in about **37% of chromophobe carcinomas** — the scar does not discriminate (Fig. 6.8). Both tumours are usually peripheral and hypovascular relative to cortex. Necrosis is T2-high and T1-low, like the oncocytoma scar.
- **CSI**: about 59% of ccRCCs contain intracytoplasmic lipid. **Once angiomyolipoma (AML) has been excluded** (macroscopic fat on standard sequences, Fig. 6.9), **an SI drop of 25% or more is diagnostic of ccRCC** among renal tumours. CSI is **highly specific but insensitive** for ccRCC.
- **Trap**: RCC rarely contains macroscopic fat; **fat plus calcification favours RCC over AML.**
- **Enhancement**: a **relative SI increase of 15% or more** on gadolinium-enhanced T1 counts as enhancement, separating cyst from RCC with 100% sensitivity; **peak at 2–4 minutes**. ccRCC enhances by **more than 200% in all three phases**, significantly more than chromophobe and papillary carcinoma.

Figure legends

- **Figure 6.6**: right 3.5 cm ccRCC with diffuse OP drop-out, avid heterogeneous enhancement and high signal on DWI at b-1000; a left subcapsular haematoma after extracorporeal shock wave lithotripsy is **T1-bright on fat-suppressed imaging and does not enhance.**
- **Figure 6.7**: pRCC — T2 heterogeneous but mainly low; isointense to muscle on T1; no significant corticomedullary enhancement, minimal central nephrographic enhancement; **low signal on the ADC map.**
- **Figure 6.8**: chromophobe carcinoma — central high T2 signal, central low T1 signal, peripheral enhancement with a non-enhancing centre.
- **Figure 6.9**: 2.5 cm AML — **no OP drop**, but signal loss on fat suppression (macroscopic fat).

TABLE 6.2 MRI CHARACTERISTICS OF RENAL MASSES (REBUILT)

Mass	Enhancement: corticomedullary / nephrographic / excretory	ADC ($\times 10^{-6}$ mm ² /s)	T2 signal
Clear cell	Avid: 230 / 250 / 227%	1698 (high)	High, heterogeneous
Papillary	Weak: 49 / 92 / 88%	884 (lowest RCC)	Low, homogeneous
Chromophobe	Intermediate: 98 / 183 / 159%	1135	High for central scar
Oncocytoma	Avid: 208 / 265 / 237%	—	High for central scar
AML	353 / 285 / 222%	—	Variable
Renal parenchyma	—	2303	—
Transitional cell carcinoma (TCC)	—	<450	High SI

Data attributed in the source to Varghese JC, Hahn PF, Clin Radiol 1997.

- **What the table is for**: the renal mass is subtyped by **three independent axes** — **vascularity, cellularity (ADC) and T2 signal**. Clear cell sits at the high end of all three; papillary at the low end.
- **AML is the only mass whose signal falls from the corticomedullary to the nephrographic phase.**
- **Trap**: overlap is large, so no firm cut-offs can be set, and **oncocytoma could not be distinguished from ccRCC.**

Evaluating Local Tumour Extent

- **Venous tumour thrombus:** MRI was historically superior to CT for renal vein and inferior vena cava thrombus; **current MRI and CT perform equally. Gadolinium separates tumour thrombus (enhances) from bland thrombus (does not).**
- **Lymph nodes:** MRI and CT both detect lymphadenopathy. **Superparamagnetic iron oxide nanoparticles** are under investigation: **normal nodes phagocytose them and lose T2 signal**; metastatic nodes do not.
- Multiparametric MRI adds information on the **tumour microenvironment**.

Concept — Enhancement is the single most important sign of malignancy in a renal mass, and MRI sees it through calcium where CT may not. Subtyping follows one pattern: clear cell is bright, hypervascular, CSI-positive and high-ADC; papillary is dark, hypovascular and low-ADC.

UPPER TRACT AND LOWER TRACT MRI FOR TRANSITIONAL CELL CARCINOMA

Upper tract

- **Magnetic resonance urography (MRU)** is added to renal mass protocols and is useful when other modalities are contraindicated. **Static MRU:** heavily T2-weighted, urine bright. **Excretory MRU:** gadolinium-enhanced T1.
- **MRU and CT urography are equally accurate for obstruction.**
- **Stones and calcification have no signal and appear as voids.** Tumour, clot, gas and sloughed papillae appear as low signal or voids against bright urine on T2.
- **Fast acquisitions (about 2 seconds per image)** separate true pathology from ureteric peristalsis.
- **Normalised ADC $\leq 451 \times 10^{-6} \text{ mm}^2/\text{s}$ separates TCC of the renal pelvis from central RCC.** ADC varies with b-value and TE, so **cut-offs need institutional validation.**
- On DWI, upper tract TCC is **hyperintense** and **homogeneous**; RCC is heterogeneous.

Bladder

- MRI beats CT through **greater contrast between bladder wall layers**, separating invasive from superficial disease with good accuracy (Fig. 6.10).
- **The detrusor is the low-signal layer on T2;** urine, mucosa and submucosa are all high signal and cannot be separated. **Muscle-invasive tumour is high signal disrupting the low-signal muscle layer.**
- Bladder TCC is **DWI-hyperintense** against tissue and urine. **Underdistension causes false negatives.**
- **Vesical Imaging-Reporting and Data System (VI-RADS):** a **5-point score** for muscle invasion from T2, DWI and DCE findings.
- **Figure 6.10:** a T2-low right wall filling defect; an enhancing polypoid mass on early DCE without wall invasion; transurethral resection of bladder tumour confirmed high-grade papillary urothelial carcinoma **without muscle invasion**.

Concept — Urine is the contrast agent on T2, so anything within it — tumour, clot, stone — is dark; in the bladder wall the question is whether the dark detrusor line is intact, and VI-RADS formalises that.

PROSTATE MAGNETIC RESONANCE IMAGING

Evolution and landmark evidence

- Prostate cancer (CaP) long lacked reliable imaging; the standard was transrectal ultrasound with an **extended sextant (12–14 core) biopsy**. Advances included **1.5 T → 3 T magnets** and better protocols.
- **PROMIS (Prostate MR Imaging Study, 2017):** level 1b evidence for **MRI before biopsy** in men with elevated prostate-specific antigen (PSA); a follow-up analysis showed **cost-effectiveness**.

- **Göteborg (Swedish screening study):** MRI in a PSA-based screening protocol (threshold 2.8 ng/mL) roughly halved biopsies.
- **PRECISION (2018):** in biopsy-naïve men, **MRI with or without targeted biopsy detected more clinically significant and less clinically insignificant cancer.**
- **Cochrane review (2019):** MRI sensitivity **0.91**, specificity **0.37** for Grade Group (GG) 2 or higher — excellent at ruling out, poor at ruling in.
- **American Urological Association (AUA) and European Association of Urology (EAU)** both include prostate MRI in the work-up of elevated PSA.

Coils

- "The currency of MRI is signal." Signal was historically boosted with surface coils and an **endorectal coil (ERC)**, at the cost of discomfort and expense.
- **At 3 T the ERC is largely historical:** image quality and detection of GG3 or higher are equivalent without it; GG2 detection is somewhat inferior. **The ERC remains useful on older 1.5 T scanners.** If used, give an **antispasmodic** to reduce rectal motion artefact.

T2-Weighted Imaging

- The **anatomic sequence**; should be **triplanar** (axial, coronal, sagittal) or equivalent (Fig. 6.11).
- **Normal peripheral zone (PZ) is high signal.** The central gland with benign prostatic hyperplasia (BPH) shows well-demarcated nodules of heterogeneous signal.
- **Low T2 signal is non-specific:** cancer, prostatitis, atrophy, scar, post-biopsy haemorrhage and BPH nodules. **Stromal BPH nodules within the PZ cause false positives.**
- **T2-weighted imaging (T2WI) alone is specific but insensitive;** adding functional sequences raises negative (NPV) and positive predictive value (PPV) **above 90%.**
- **T2WI assesses extraprostatic extension (EPE) and seminal vesicle invasion (SVI),** both appearing as low signal.
- **Figure 6.11:** a well-circumscribed heterogeneous BPH nodule within the PZ, with restriction on ADC and type 2–3 enhancement curves, was **benign on fusion biopsy.**

Diffusion-Weighted Imaging/Apparent Diffusion Coefficient Maps

- DWI measures **Brownian motion of water.** Higher cellularity → tighter packing → restricted water → **high signal on DWI.**
- **b-values** set the threshold for detecting restriction; as the b-value rises, less-restricted tissue loses signal. Use multiple b-values, **including at least one above 1000;** some add **b-2000**, on which cancer is bright relative to the gland.
- The **ADC map** is the quantitative counterpart; **cancer is dark (low signal) on ADC.**
- **ADC correlates inversely with Gleason score:** higher cellularity → less extracellular space → more restriction → lower ADC.
- **Figure 6.12:** a large anterior central gland GG 4+4 cancer after 16 negative biopsies — low ADC, bright b-2000, type 3 curve, elevated transfer constant and rate constant, found on MR-ultrasound fusion biopsy.

Dynamic Contrast-Enhanced MRI

- T1-weighted imaging with GBCAs measuring **perfusion and vascular permeability** over time — not anatomy, and not simply "enhances or not."

- **Temporal resolution ≤ 15 seconds.** Under 10 seconds was once preferred, but no benefit was shown across 1.4–14.9 seconds.
- Three levels of analysis: **qualitative** (visual early enhancement and washout); **semi-quantitative** (time-intensity curves); **quantitative** (Tofts pharmacokinetic model).
- **Transfer constant (K-trans):** rate of transfer from the intravascular to the extracellular space (permeability and flow). **Rate constant (K-ep):** rate of efflux back to the vascular space. Not yet in routine workflow; being studied for decision-support software.
- Sensitivity and specificity vary widely with patient selection, technique, pathology correlation and reader experience. **DCE is secondary to T2WI and DWI in scoring.**

DYNAMIC CONTRAST-ENHANCED TIME-INTENSITY CURVES (FIG. 6.13)

Curve	Pattern	Meaning
Type 1	Persistent progressive rise	Normal
Type 2	Early enhancement, then plateau (no washout)	Indeterminate
Type 3	Early enhancement, then early washout	Most suggestive of CaP ; overlaps with inflammation

- **Because all three curves overlap with benign conditions,** the approaches are combined within multiparametric MRI.

Multiparametric Magnetic Resonance Imaging

- **Anatomic:** T1 and T2. **Functional:** DWI with ADC, DCE, and optionally spectroscopy.
- **MR spectroscopy is not routine:** about **15 minutes**, labour-intensive, and seldom changes interpretation.
- **T2 + DWI + DCE gives an NPV above 90%.**
- Quality needs magnet tuning, dedicated staff and pathology feedback; start from the ACR's **Prostate Imaging-Reporting and Data System (PI-RADS) v2.1**. The technologist reviews images during the scan and repeats sequences as needed.
- **Uses:** detection, quantification, staging, treatment planning; **selection for active surveillance** (good NPV); bone and nodal assessment at diagnosis (moderate-to-good nodal sensitivity and specificity).

Biparametric Magnetic Resonance Imaging

- **T2 + DWI, without DCE:** lower cost, shorter scan, **no GBCA risk**, but may limit diagnostic capability and **limits assessment beyond the prostate** (staging).
- Accuracy varies with magnet strength and reader experience; a meta-analysis found **no significant difference from mpMRI for CaP detection**.
- **Reasonable for screening low-risk men** when image quality is optimised and experienced readers interpret.

MULTIPARAMETRIC VERSUS BIPARAMETRIC MRI

	Multiparametric (mpMRI)	Biparametric (bpMRI)
Sequences	T2 + DWI + DCE	T2 + DWI
Contrast	Gadolinium	None
Cost and time	Higher	Lower
Detection of CaP	Standard	No significant difference
Staging beyond prostate	Better	Limited

	Multiparametric (mpMRI)	Biparametric (bpMRI)
Best role	Standard of care	Low-risk screening, expert readers

Quality Control for Prostate Magnetic Resonance Imaging

- Rectal gas, patient motion and metal implants degrade images. **Prostate Imaging Quality (PI-QUAL)** assesses whether lesions can be ruled in or out given image quality: axial T2 matching DWI and DCE, clear capsular border and seminal vesicles, among other criteria.

TABLE 6.3 PI-QUAL SCORING SHEET

Score	Criteria	Rule in significant lesions	Rule out significant lesions
1	All sequences below diagnostic standard	No	No
2	Only one sequence acceptable	No	No
3	At least two sequences acceptable together	Yes	No
4	Two or more sequences independently optimal	Yes	Yes
5	All sequences optimal	Yes	Yes

From Giganti F, Allen C, Emberton M, et al: PI-QUAL, from the PRECISION trial. Eur Urol Oncol 2020.

- At scores 1–2, reports should not include a PI-RADS or Likert score. Score 3 is the trap: it can rule in but not rule out.
- Figure 6.14 shows an excellent-quality scan with a right mid-gland anterior PZ lesion.

Reporting: PI-RADS v2.1 and PRECISE

- Prostate MRI benefits from **consensus reading, pathology correlation and designated experienced readers**. The PI-RADS report gives **zone, sector and axial T2 slice number**.

TABLE 6.4 PI-RADS V2.1 ASSESSMENT CATEGORIES AND TABLE 6.5 PRECISE SCORING SYSTEM

Score	PI-RADS v2.1 — likelihood of clinically significant disease	PRECISE — likelihood of radiological progression (active surveillance)
1	Very low — highly unlikely	Resolution of previously suspicious features
2	Low — unlikely	Reduction in volume or conspicuity
3	Intermediate — equivocal	Stable; no new focal or diffuse lesion
4	High — likely	Significant increase in size or conspicuity
5	Very high — highly likely	Definite radiological stage progression (EPE, SVI, nodes, metastasis)

PI-RADS from Weinreb JC et al, ACR 2019; PRECISE (Prostate Cancer Radiological Estimation of Change in Sequential Evaluation) from Moore CM et al, Eur Urol 2017.

- PI-RADS scores a single scan; PRECISE scores change between scans on active surveillance. Both run 1–5, and PRECISE 3 means stability, not equivocation.
- Figure 6.15 (sector map):** at base, mid-gland and apex — PZ anterior, posterolateral and posteromedial; transition zone (TZ) anterior and posterior; central zone; anterior fibromuscular stroma; urethral sphincter; seminal vesicles.

Magnetic Resonance Fusion-Guided Prostate Biopsy

- The dominant determinant of performance is mpMRI quality and the imaging team, not the fusion platform.
- Targeted biopsy or mpMRI alone is not yet an alternative to the standard of care in screening.

- mpMRI helps select men with elevated PSA for biopsy (PSA rises with inflammation, infection, BPH, manipulation); it can show BPH or inflammation and may **spare a repeat biopsy** after a negative 12-core biopsy.
- **mpMRI detects intermediate- and high-risk disease and misses low-grade, low-volume cancer — desirable, as it reduces overdiagnosis and overtreatment.** Over a third of men with elevated PSA have no visible lesion, and 12-core biopsy then rarely finds clinically significant cancer.
- **Göteborg-2 (2022): targeted biopsy alone** avoided overdetection of insignificant cancer and **missed under 0.2% of intermediate-risk CaP.**
- Fusion reaches **anterior and central gland lesions** undersampled by systematic biopsy; 12-core biopsy also misses many PZ lesions.
- **Fusion outperforms cognitive targeting:** a randomised trial showed higher overall and clinically significant detection with fusion plus template than cognitive plus template biopsy.
- **Cancer detection rate for fusion-targeted biopsy of moderate- to high-suspicion lesions: about 50–72%.** Deep-learning predictions with PI-RADS and clinical variables are being developed into risk nomograms.

Concept — Prostate MRI is a filter before the needle: T2 gives anatomy, DWI and ADC give cellularity and drive the score, and DCE is the tiebreaker. Quality (PI-QUAL) must be assured before scoring (PI-RADS), change is tracked with PRECISE, and the yield of targeting depends on the scan and the reader rather than the device.

KEY POINTS

- **NSF is now largely historical;** newer GBCAs can be used safely in severe renal insufficiency.
- **Pheochromocytoma, adrenal metastasis and primary ACC are all bright on T2.**
- In a renal mass on MRI, **enhancement is the most important indicator of malignancy.**
- Prostate MRI has **transformed the CaP screening and diagnostic pathway** and improved cancer detection.